

Unified Deep Learning Optimization, Initialization & Scaling Cheat Sheet

1 Notation

- Parameters $w \in \mathbb{R}^d$, layer weights $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$
- Loss $L(w)$, sample loss $\ell_i(w)$, gradient $g_t = \nabla L(w_t)$
- SVD: $A = U\Sigma V^\top$; fan-in = d_{in} , fan-out = d_{out}
- RMS norm: $\|x\|_{\text{RMS}} = \sqrt{\frac{1}{d} \sum_i x_i^2}$

2 First-Order Optimization

2.1 Gradient Descent (GD)

$$w_{t+1} = w_t - \eta g_t.$$

Stability assumptions: For quadratic loss, $\eta \leq 1/\lambda_{\max}(X^\top X)$. For general smooth L , require $\eta < 2/L$ (Lipschitz gradient constant L).

Geometry (linear models): Updates lie in $\text{col}(X)$; convergence to minimum-norm interpolator requires $w_0 \in \ker(X)$, stable η , and zero-loss interpolation.

Implicit bias: Small-singular-value directions updated slowly $\Rightarrow$ ridge-like behavior.

2.2 Stochastic Gradient Descent (SGD)

$$w_{t+1} = w_t - \eta g_{i_t}.$$

Noise aids saddle escape; convergence via Robbins–Monro or constant LR in interpolation. In NNs, updates lie in span of backpropagated feature gradients.

2.3 Momentum

$$v_{t+1} = \beta v_t + (1 - \beta)g_t, \quad w_{t+1} = w_t - \eta v_{t+1}.$$

Eigen-dynamics for Hessian eigenvalue λ : $|1 - \eta\lambda + \beta|$. Momentum accelerates but does *not* fix conditioning.

2.4 Adam / AdamW

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1)g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2)g_t^2,$$
$$w_{t+1} = w_t - \eta \frac{m_t / (1 - \beta_1^t)}{\sqrt{v_t / (1 - \beta_2^t)} + \epsilon}.$$

AdamW decouples weight decay; adaptive scaling improves conditioning but may generalize worse than SGD.

2.5 Sign-SGD

$$\min_{\|\Delta w\|_\infty \leq \eta} \langle g_t, \Delta w \rangle \Rightarrow \Delta w = -\eta \text{sign}(g_t).$$

Illustrates how *choice of update norm* defines the optimizer's geometry.

3 Ridge Regression & Implicit Regularization

$$w_\lambda = (X^\top X + \lambda I)^{-1} X^\top y = \sum_i \frac{\sigma_i}{\sigma_i^2 + \lambda} u_i (v_i^\top y).$$

Gradient flow = exact ridge; discrete GD approximates ridge for small η . Early stopping damps small- σ components.

4 NTK, Linearization, and Learning Regimes

4.1 Linearization

$$f(\theta + \Delta\theta) \approx f(\theta) + J(\theta)\Delta\theta, \quad J = \nabla_{\theta} f.$$

Valid when

$$\|\Delta f\| = \|J\Delta\theta\| \ll 1, \quad \|\theta_t - \theta_0\|_2 \ll \frac{1}{\sqrt{\text{width}}}.$$

4.2 Neural Tangent Kernel

$$K(x, x') = J(x, \theta_0)^{\top} J(x', \theta_0).$$

Under standard parametrization and $\text{width} \rightarrow \infty$, parameters move little $\Rightarrow$ NTK $\approx$ constant $\Rightarrow$ kernel regression (lazy training).

4.3 Wide Nets Do Not Automatically Enter NTK

NTK regime arises only if learning-rate/parameterization scales so that $\|\Delta f\| \ll 1$. Xavier/He ensure stable activations, not lazy training by themselves.

5 Initialization (Variance Preservation)

5.1 Xavier

$$w_{ij} \sim \mathcal{N}\left(0, \frac{1}{d_{\text{in}}}\right).$$

5.2 He (ReLU)

$$w_{ij} \sim \mathcal{N}\left(0, \frac{2}{d_{\text{in}}}\right).$$

Goal: keep activations/gradients $\mathcal{O}(1)$; LR scaling (not initialization alone) determines NTK vs. feature learning.

6 Gradient Scaling With Width (Key for μP)

Elementwise: $G_{ij} = \mathcal{O}(1/\sqrt{d_{\text{in}}})$. Frobenius norm:

$$\|G\|_F = \mathcal{O}\left(\sqrt{\frac{d_{\text{out}}}{d_{\text{in}}}}\right).$$

Thus widening layers changes functional update size unless LR is scaled.

7 RMS $\rightarrow$ RMS Norm (Initialization–Optimizer Alignment)

$$\|x\|_{\text{RMS}} = \sqrt{\frac{1}{d} \sum x_i^2}, \quad \|A\|_{\text{RMS} \rightarrow \text{RMS}} = \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}} \|A\|_2.$$

(Identity depends on RMS definition used in slides.) Provides correct scaling to keep $\|\Delta f\|$ invariant across width.

8 μP : Maximal Update Parameterization

8.1 Core Principle

Scale LR and initialization so that functional updates remain $\mathcal{O}(1)$ as width grows.

8.2 Conditions

$$\|\Delta W_{\ell}\|_{\text{RMS} \rightarrow \text{RMS}} = \mathcal{O}(1), \quad \text{activations} = \mathcal{O}(1), \quad \text{feature updates} \neq 0.$$

8.3 Scaling Rules (simplified)

$$\eta_\ell \propto \frac{1}{d_{\text{in}}}, \quad W_\ell^{\text{init}} \propto \frac{1}{\sqrt{d_{\text{in}}}}.$$

Ensures LR transferability and prevents NTK collapse.

9 Second-Order Geometry Approximations

9.1 Shampoo

$$W_{t+1} = W_t - \eta(G_t G_t^\top + \epsilon I)^{-1/2} G_t (G_t^\top G_t + \epsilon I)^{-1/2}.$$

Approximates curvature using Kronecker factors; requires ϵI for stability.

9.2 Muon (Momentum + Direction Normalization)

$$B_t = \mu B_{t-1} + G_t.$$

Rank-1 structure in minibatch-1 enables efficient SVD-like operations. Normalize via Newton–Schulz:

$$B_t = U \Sigma V^\top \mapsto UV^\top.$$

Update:

$$W_{t+1} = W_t - \eta \tilde{O}_t, \quad \tilde{O}_t \approx UV^\top.$$

Muon normalizes *gradient* singular values (not Hessian curvature), stabilizing geometry across width.

10 Unifying Principles

1. **Optimization = geometry**: each optimizer solves a constrained first-order step under a chosen norm.
2. **Initialization sets scale; optimizer must respect it**: Xavier/He stabilize signals; μ P/Muon stabilize functional updates.
3. **NTK emerges from small functional movement**: lazy regime $\Leftrightarrow \|\Delta f\| \ll 1$, controlled by LR scaling.
4. **Width determines learning regime**: standard scaling $\rightarrow$ NTK; μ P rescales updates to enable feature learning.
5. **Curvature + noise govern dynamics**: SGD noise, momentum filtering, and geometry-aware preconditioners shape optimization.
6. **Hyperparameter transferability requires scaling**: μ P ensures LR/initialization stability across width.